AN ADJOINT PAIR TRANSFERS FUNCTORIAL INJECTIVE EMBEDDINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, with the construction displayed; the closing sentence 'Compare with the proof of Lemma ...' points to the argument just adapted rather than declining a step, and it is preserved verbatim.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Functorial injective embeddings. Same chapter, the definition of a functorial injective embedding. Supplies the data A to (A -> J(A)) on the source category that the proof transports to the target.
- Transfer of enough injectives across an adjoint pair. Same chapter, Lemma lemma-adjoint-enough-injectives. Supplies the argument being adapted, in which an injection of the image of an object into an injective object is transported across the adjunction.

Source reference. Original lemma 0160: sources/stacks-homology/pinned-original-source.tex, source lines 7546–7546. Navigation pointer invoked at proof line 7573; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \to \mathcal{B}$ and $v : \mathcal{B} \to \mathcal{A}$ be additive functors. Assume*

(1) *u is right adjoint to v,*
(2) *v transforms injective maps into injective maps,*
(3) *$\mathcal{A}$ has enough injectives,*

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(4) $vB = 0$ *implies* $B = 0$ *for any* $B \in \mathrm{Ob}(\mathcal{B})$, *and*
(5) $\mathcal{A}$ *has functorial injective embeddings.*

Then $\mathcal{B}$ *has functorial injective embeddings.*

Proof. Let $A \mapsto (A \to J(A))$ be a functorial injective embedding on $\mathcal{A}$. Then $B \mapsto (B \to uJ(vB))$ is a functorial injective embedding on $\mathcal{B}$. Compare with the proof of Lemma 0160. □